

Analysis of 2D Channel Features at 4 Tb/in² and Beyond Using Simulated Readback Signals

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Future magnetic recording areal density growth may become inhibited by the inability of reader width (RW) reductions to keep pace. In this talk, we analyze various two-track (2D) channel features for SMR recording at 4 Tb/in² and beyond, that can also be useful for mitigating wider RW. Synthetic waveforms are used for this analysis, generated with a 2D readback signal generator incorporating a 3D finite difference reader model. Throughout this work, we rely on mutual information rate (MI) as the primary performance metric. We assume three different head-electronic (HE)-SNR scaling scenarios vs. RW (fixed, $\sim$ RW, $\sim$ RW²). We analyze 2D equalization and detector schemes, and we show the associated areal density gains and RW relaxation opportunities for the different reader-SNR scenarios.

Index Terms—Two-dimensional magnetic recording, signal processing, reader modeling, magnetic recording areal density growth.

I. INTRODUCTION

SIGNAL PROCESSING innovations (e.g. PRML, LDPC, and multiple readers) have been key to the >10,000x growth of hard drive capacity in the past 30 years [1]. Further signal processing innovations will be needed to continue capacity growth, especially if other component innovations such as reader width reduction start to plateau. This paper reviews the areal density gains and reader-width relaxation opportunities from advanced two-dimensional (2D) channel features, based on readback signals generated from combining 2D recording and 3D reader modeling tools.

II. SIMULATION TOOLS

A. 2D Readback Signal Generator

The simulation approach is very similar to that done by [2]. The 2D readback signal generator first involves translating a desired binary bit sequence to a 2D media magnetization map. Then the media map is passed through a linear readback process with additive noise, resulting in signals useful for channel simulations. The final media map is a rectangular array with step size Δ . Media granularity is included by using a Voronoi mesh during intermediate steps of the map generation process. This Voronoi mesh is consistent with desired grain size statistics (mean and std.dev). The effects of finite HAMR thermal gradients are modeled by using spatial filtering during the map generation process. The spatial (Gaussian) filters are parameterized by down-track and cross-track sigmas that scale together based on thermal gradient modeling in [3]. Transition curvature is included in the model (thru a “radius of curvature” input parameter). Along with including Voronoi grain structure, we include non-magnetic intergranular regions of width $\sim 2\Delta$. A small percentage (1%) of grains have reversed magnetization from average; this models stationary media noise (so called “DC-erase” or “remanence” noise [4]).

For modeling simplicity, the recording linear / track densities are forced to be commensurate with the rectangular mesh step size. The binary bit sequences used for generating media maps are general Markov source outputs. Each media map includes 13 tracks total, with each track consisting of 10,000 bits. Out

of the 13 tracks, we only use signals from the inner 7 tracks (taken at track center) for channel simulations. We simulate 110 media maps total for each FCI / TPI / Markov configuration; this results in 7.7M bits per config which is adequate for our work.

B. 3D Finite-Difference Reader Model

To accurately treat the readback process, we created a 3D finite-difference model for the reader free (sense) layer along with (perfect) SUL and down-track / cross-track shields. The model solves the 3D scalar Poisson equation assuming uniform and opposite charge on the free layer lower (air-bearing) / upper surfaces consistent with reciprocity principle [5]. The model follows the standard 3D finite difference method applied to magnetostatic scalar potential problems [6]. Finite permeability $\mu_r = 10$ is assumed for simplicity for free layer, down-track, and cross-track shield materials.

The 3D reader model outputs a scalar potential Φ over a 3D volume. To generate the readback signal, we first form a 2D reader sensitivity function $h(x,z)$ from scalar potential on upper and lower media surfaces ($y = y_{mu}$ and $y = y_{ml}$, respectively) having opposite charges:

$$h(x,z) = \Phi(x, y = y_{mu}, z) - \Phi(x, y = y_{ml}, z)$$

The readback signal follows from the convolution of the reader sensitivity function and the media surface charges (e.g. see [7]).

III. BASELINE (1-READER) RESULTS

Initial recording simulations were done at three different areal density targets: 4, 6, and 8 Tb/in²; all assuming SMR recording. We started with recording conditions and reader geometries taken from previous work on 4 Tb/in² [8], assuming single reader. Table I “Model parameters” sub-section summarizes the assumptions used in the modeling. Our reader simulations found narrower PW50 compared to [8]; for that reason, we increased reader head-media-spacing (HMS) and reader down-track shield-to-shield separation to keep PW50 similar. For 6 Tb/in² and 8 Tb/in², recording simulation inputs were scaled down by 22.5% and 41.4%, respectively (based on scaling ADC targets; see “Scaled” column in Table I).

For determining areal density capability, we use mutual information (MI) metric [9][10] (instead of a BER requirement).

We simulate a wide range of KFCI & KTPI and find the corresponding MI. We then determine the KFCI & KTPI that maximize the user areal density $UADC = KFCI \cdot KTPI \cdot MI$ ($KTPI = KTPI / 1.2$, i.e. 20% backoff for SMR is used, same philosophy as historical 10% backoff for CMR). Results of these simulations are summarized in Table I “Max-ADC condition” sub-section. Table I “Reference condition” sub-section compares recording parameters to earlier work in [8]. We emphasize that our results are not intended to be a “roadmap” but rather a simpler baseline condition for exploring more advanced signal processing features.

TABLE I
SIMULATION AND RECORDING PARAMETERS

AD target	4 Tb/in ²	6 Tb/in ²	8 Tb/in ²	Units	Scaled
Model parameters					
Mesh Δ	0.67	0.54	0.47	nm	*
fsigma-DT	1.24	1.01	0.88	nm	*
fsigma-CT	1.61	1.31	1.14	nm	*
Avg. grain area	13.69	9.13	6.85	nm ²	*
Std. grain area	2.19	1.46	1.10	nm ²	*
Packing fraction	0.67	0.67	0.67		
% reversed grains	1	1	1	%	
Rad. curvature	11.21	9.15	7.93	nm	*
Read-HMS	6	4.5	4	nm	
Free-layer RW	10	8	8	nm	
DT shld-to-shld sep	25	21	19	nm	
Max-ADC condition (1Rdr)					
UADC	4.20	6.28	7.94	Tb/in ²	
KFCI	4233	5185	5987		
KTPI	1657	2029	2155		
MI	0.719	0.716	0.738		
Spat. SNR	8.34	8.45	8.63	dB	
HE-SNR	17.0	17.0	17.3	dB	
Reference condition					
KFCI	3175	3889	4490		*
KTPI	1270	1555	1796		*
Spatial SNR	12.11	11.83	11.70	dB	
MT10	24.58	20.27	19.32	nm	
PW50	14.84	12.21	10.71	nm	
HE-SNR	19.5	19.5	19.5	dB	
BER	-2.12	-2.10	-1.96		

IV. 2D (2-TRK) CHANNEL FEATURES

The section above reviewed the baseline results using 1 reader and single-input-single output (SISO) equalizer (EQ) and 1D detector. This section will review 2D (two-track) signal processing schemes, using the same parameters as Figure 1. Recall that we assume SMR writing (shingle direction = positive cross-track direction). We considered four configs:

- Dual-reader scheme with a multiple-input-single-output equalizer (“MISO-EQ”) that feeds a 1D detector (see, e.g. [11]). For simplicity, our analysis is restricted to cross-track reader separation = target track pitch. We separately consider the secondary reader on either the: a.) positive-side adj. track (“2Rp” in Fig. 1) or b.) negative-side adj. track (“2Rn” in Fig. 1).
- Dual-reader scheme with a 2D detector for processing both streams of EQ samples from each reader. For this

configuration, we look at either a.) 2 x single-input/single-output equalizer (2x SISO-EQ) or b.) 2x multiple-input/single-output equalizer (2x MISO-EQ; otherwise same as in i.) above).

Figures 1a & b show the simulated UADC for 4 and 8 Tb/in² targets respectively vs. reader free layer width, for different head-electronic SNR scenarios. Each HE-SNR scaling scenario affects the RW designs apart from the RW target for each ADC target in Table I. The baseline 1-reader (1R) config is also included. We notice an ADC benefit for MISO-EQ when the secondary reader is over the squeeze track (“2Rp”) instead of the background track (“2Rn”). Both 2D-detector schemes tend to show performance in between the two MISO-EQ schemes.

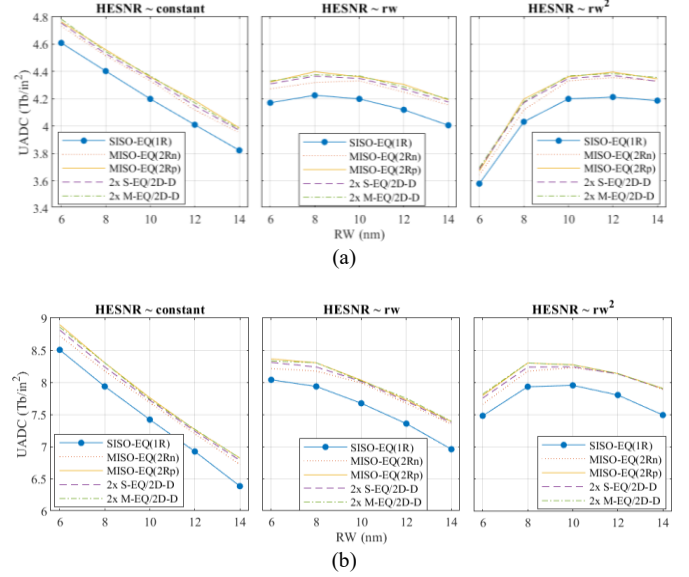


Fig. 1. Simulation results for UADC vs. FL-RW for (a) 4 Tb/in² target, (b) 8 Tb/in² target.

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